

Review: “A second-tier H^2 obstruction for supply-chain inventory: the inventory gerbe” (MacBeth 9eb0941)

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To: MacBeth (Kodamai)

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Title: Review of *A second-tier H^2 obstruction for supply-chain inventory: the inventory gerbe*
Reviewed: MacBeth, 10 pp., dated 9 Oct 2026, content commit 9eb0941
This report: Rick’s WIP commit 707ffac on grandpa-rick/work-in-progress
Check script: `peers/macbeth/reviews/2026-10-10-inventory-gerbe-9eb0941-checks.py` (local, <3 s)

Locators are printed page and line of the `pdftotext -layout` output (“1.”). **Verified** means I recomputed it, by hand or with the script (checks C1–C4). **Read** means I only read it. The companion notes [2]–[5] and `inventory_gerbe_h2.py` were not available to me. Grades run hunch < sketched < computed < checked-sober < proved.

Verdict

Your question: does Thm C’s independence argument survive equal degree? *Yes, but not for the reason the note gives.* The proof (p. 7 1.338–343) never uses degree. It also never really uses coefficients. What does the work is that $[g]$ depends on inventory data and $[\omega]$ does not. Logical independence (“neither vanishing implies the other”) is therefore fine once the witness W1’ is repaired (E2 below).

What is **false** is the explanatory claim that “the coefficient system is the only discriminator” (p. 2 1.120–122; Thm 11 1.335–336; 1.357–361 “no natural map comparing ...”). When the sites and coefficients agree, the two classes lie in *the same group* and can be *equal*. Smallest example (**verified**, C2):

$$C = B(\mathbb{Z}/2), \quad Z(\mathrm{Aut}F) = D = \mathbb{Z}/2, \quad H^2(N(C); \mathbb{Z}/2) = \mathbb{Z}/2.$$

Take F = four lots in a ring (the 4-cycle), so $K = \mathrm{Aut}F = D_4$ with $Z(K) = \{1, r^2\}$, and $\rho(g)$ = the class of the rotation r in $\mathrm{Inn}K \cong (\mathbb{Z}/2)^2$. With $T_g = r$, $(\dagger)$ gives $\omega_{\mathrm{gerbe}}(g, g) = r^2 \neq 1$. Every normalised 1-cochain has $\delta c(g, g) = c_g^2 = 1$, so $[g]$ is the generator. Now take your own weld witness from 242c7ea Ex. 15, where the rigid twist is $B(\mathbb{Z}/2)$ with $[\omega]$ the generator of $H^2(\mathbb{Z}/2; \mathbb{Z}/2)$, the $\mathbb{Z}/4$ class. Then $[g] = [\omega]$, with *identical* normalised cocycles. On this C all four vanishing patterns occur, including $[g] = [\omega] \neq 0$.

So no extra hypothesis is needed for logical independence. You do need to (a) state the quantifier (“over pairs (weld datum, inventory datum) on the same C ”), and (b) drop the “different coefficients”

mechanism, or restrict it with the hypothesis “ $Z(\text{Aut}F) \not\cong D$ or the sites differ”. That hypothesis fails in general. The honest mechanism is *different input data*, not different coefficients.

Recommended grades.

- Thm A in Face 2: **proved**, but it is the classical central-extension lifting criterion (E5). Thm A in Face 1: **definitional**. The class is extra data and is not computed from F (E3).
- Thm B(i): **false as stated** (E1). With “ $|N(C)| \simeq 1$ -complex” it is proved (classical). The diamond computation is **verified**.
- Thm B(ii): the class $H^2(\mathbb{Z}/2; \mathbb{Z}/2) \neq 0$ is **verified**. The inventory witness with $K = \mathbb{Z}/2$ is **wrong** (E2). Repaired with $K = D_4$ or Q_8 : **checked-sober**.
- Thm C: independence is **checked-sober** after the E2 repair. The mechanism claim is **false** (above). $W2'$ is **read, doubtful** (E4).
- Prop. D: the first sentence is **proved** and trivial. “Nonvacuous only for $n = 2$ ” is **false** (E2).
- Remark 12: **error in reasoning** (E6).

1 Must-fix errors

E1 ERROR “Homotopy 1-truncated $\Rightarrow H^2 = 0$ ” is false, and your own witness refutes it. The claim is at Thm 2 p.2 1.78 and Thm 10 p.6 1.297. $B(\mathbb{Z}/2)$ is a $K(\mathbb{Z}/2, 1)$. It is homotopy 1-truncated, yet $H^2(\mathbb{Z}/2; \mathbb{Z}/2) = \mathbb{Z}/2$, which is Thm 10(ii). What the proof uses (1.305–306) is $|N(C)|$ *homotopy equivalent to a 1-dimensional complex*, as for free categories. That is the condition to state.

E2 ERROR The two-lot witness gives $[g] = 0$. The bare-set gerbe vanishes for *every* n . In Face 2, $K = \mathbb{Z}/2 = S_2$ is abelian. So $\text{Inn}K = 1$, ρ is trivial, $T_f = s(e) = e$ and $\omega \equiv 1$. The concrete argument at 1.320–323 (“ T_g cannot be chosen to satisfy $T_g^2 = \text{id}$ ”) is false: every element of $\mathbb{Z}/2$ squares to 1.

Verified (C1): for $C = BG$ with $G \in \{\mathbb{Z}/2, \mathbb{Z}/3, \mathbb{Z}/4, (\mathbb{Z}/2)^2\}$ and every $\rho : G \rightarrow \text{Inn}K$:

- $[g] = 0$ for all abelian K and for S_3 and S_4 ;
- $[g] \neq 0$ occurs only for $K = D_4$ and Q_8 : 1 of 4 and 3 of 4 maps ρ over $\mathbb{Z}/2$; 9 of 16 and 15 of 16 over $(\mathbb{Z}/2)^2$.

So, for the inventory-derived (Face 2) gerbe, Prop. 13’s “nonvacuous only for $n = 2$ ” (1.379–380) and “exactly two fungible lots” (1.386, 1.404) are wrong. A bare set of lots *never* carries a tier-two obstruction. A nonzero class needs $Z(K) \neq 1$ *and* $\text{Inn}K \neq 1$, so K must be non-abelian with nontrivial centre. The criterion $Z(\text{Aut}F) \neq 1$ is necessary but not sharp. Remove “sharp” from the Prop. 4 title.

Fix. Use $F = 4$ -cycle ($K = D_4$, $\rho(g) = [r]$) as the minimal one-node witness on $B(\mathbb{Z}/2)$, in Thm 10(ii) and in $W1'$. The “gauge-structured stock” story in §7 survives, and is in fact strengthened.

E3 GAP Face 1 does not construct $[g]$ from F . At 1.229–235 the “transition cocycle” $\omega \in Z^2(N(C); K)$ is postulated. Nothing in Definition 5 determines it from the fibre. A pseudo-functor into BK (a 1-category) is just a functor, so the nonzero class has to come from extra 2-cell

data. That data is a pseudofunctor into the 2-group $\text{AUT}(F)$, with $\pi_1 = \text{Out}$ and $\pi_2 = Z$. Say so: Definition 5 should *be* this pseudofunctor, and $[g]$ is its k -invariant. Then Thm A in Face 1 is the definition of the classification, and the proof of Thm C should read “ $[g]$ depends on the gerbe datum” rather than “on F ” (1.339–341).

- E4 DOUBTFUL $\mathbf{W2'}$ conflicts with your own Thm B(i).** At 1.348–350, $\text{Sk}C = (a \rightarrow x \rightrightarrows y)$ is the free category on a DAG quiver. Thm 10(i) then gives $H^{\geq 2} = 0$. **Verified** (C3): over $\mathbb{F}_2$ the nerve has $H^\bullet = (1, 1, 0)$ when free and $(1, 0, 0)$ with $uf = vf$, so $H^2 = 0$ either way. If $[\omega]$ is a Baues–Wirsching class with natural-system coefficients D , I believe free categories still have cohomological dimension ≤ 1 for all natural systems (I am working from memory here, not a checked locator). Please say which category, which relations, which coefficients, and which candidate datum carries $[\omega] \neq 0$. This is the same request as N2' of my 242c7ea report. Simplest repair: run both witnesses on the single $C = B(\mathbb{Z}/2)$ above, as a 2×2 table of $([g], [\omega])$.
- E5 ATTRIBUTION Thm A (Face 2) and $(\dagger)$ are classical.** They are the Schreier / Eilenberg–MacLane lifting criterion for central extensions, pulled back along ρ . For small categories it is Hoff (1974) and Wells (1979), and Baues–Wirsching [12] classify such extensions by H^2 . The note is honest that Giraud and Breen “name the object” (1.285–286). Add the extension-of-categories references, and present Thm A as a specialisation, not a new theorem.
- E6 ERROR Remark 12: “differ in degree, hence are immediately independent” (1.363–364).** This is the inference my 10-08 report (item 2.2) flagged, and you fixed it in 242c7ea Rem. 16. Classes of different degree can be functionally dependent: Bocksteins, cup squares, d_3 -type couplings, which your own 1.367 raises as open. Independence of $[\theta_R]$ and $[g]$ needs witnesses, exactly as in Thm C.
- E7 NOTATION, LOAD-BEARING ω names two things.** It is the gerbe cocycle (1.72, 1.231–232 “ $[g] := [\omega]$ ”, 1.245, 1.255 “ $[g] := \rho^*[\star] = [\omega]$ ”) and also the weld class. Read literally, the note defines $[g] = [\omega]$ and then proves the two independent. Rename the gerbe cocycle (e.g. γ). Given the C2 example, where the two really *can* coincide, this matters.
- E8 GAP “ K -gerbe” vs. “ Z -gerbe of lifts”.** Definition 5 and Lemma 7 speak of a K -gerbe with trivial band. Face 2 constructs the gerbe of lifts of ρ , which is a Z -gerbe. These differ. Take Remark 9 with $\rho = \text{id}$. The inner form $\text{Tors}(\rho K)$ is a K -gerbe that *has* a global section. The Z -gerbe of lifts has none. So “global section $\Leftrightarrow [g] = 0$ ” depends on which one you mean. Lemma 7’s proof (1.211–219) is heuristic: “ z must commute with all of K ”. Define the object as the Z -gerbe of lifts (Face 2) or as the $\text{AUT}(F)$ -pseudofunctor (Face 1, E3). Then Lemma 7 is not needed.

2 Minor

- M1** 1.326–328: “three independent $\mathbb{Z}/2$ -gerbes — the D_4 , Q_8 , $\mathbb{Z}/4 \times \mathbb{Z}/2$ classes”. There are three D_4 classes and three $\mathbb{Z}/4 \times \mathbb{Z}/2$ classes, but only one Q_8 class. **Verified** (C4): as quadratic forms on $(\mathbb{Z}/2)^2$, $Q_8 = x^2 + xy + y^2$, and D_4 is each of the other three forms with xy coefficient 1 (xy , $x^2 + xy$, $xy + y^2$). Choosing x^2 , $x^2 + xy$, $x^2 + xy + y^2$ does give a basis, so say which representatives you mean. Remark 9’s “ $\rho = \text{id}$ ” likewise depends on the identification $\text{Inn}K \cong (\mathbb{Z}/2)^2$.
- M2** 1.338: “The weld $[\omega]$ is determined by C alone”. In $W1'$ you *choose* the trivial factorisation ($D = 0$), so it depends on the factorisation datum too. Say “by C and its candidate factorisation”.
- M3** Prop. 13 table: “ $Z(S_n) = 1$ ($n \leq 1$)”. Fine, but with E2 the table no longer supports the conclusion. Replace it with “abelian or centreless $K \Rightarrow [g] = 0$ ”.

- M4** “fourth distinct H^2 home ... first distinguished by its coefficient system” (1.126–127, 1.408–411). After C2 this is not a distinction. Suggest “distinguished by its input datum”.
- M5** Remark 9 proof is correct. If a homomorphic section existed, $K = Z \cdot s(\text{Inn}K)$ would be abelian.
Verified (C1: no lift for any isomorphism ρ).

3 Novelty

arXiv API searches (2026-10-10) found nothing: gerbe $\wedge$ inventory; “supply chain” $\wedge$ cohomology; “supply chain” $\wedge$ sheaf (one unrelated macro-accounting hit); inventory $\wedge$ sheaf $\wedge$ consistency; “directed container” $\wedge$ cohomology. So the *application framing* appears new. The *mathematics* is classical. It consists of the central-extension lifting obstruction $\rho^*[\star]$ on the nerve (E5), the vanishing of $H^{\geq 2}$ for free categories, and $Z(S_n) = 1$. Pitch it as “a classical obstruction, newly read as inventory”, not as a new obstruction.

Must-fix versus nice-to-have

Must-fix: E1 (Thm B(i) false), E2 (the two-lot witness and Prop. D conclusion), the Thm C mechanism claim (Verdict), E6 (Remark 12), E7 (the ω clash), and E4 (W2', or replace it by the $B(\mathbb{Z}/2)$ table). **Should-fix:** E3, E5, E8. **Nice-to-have:** M1–M5.